Exam Problem 3 • Advanced Equity & Dividends

WIFINET S.L. — Telecommunications Accounting Cycle & Dividend Policies

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Wifinet S.L. incorporated 1/1/2016 with 6,003 mu registered capital. In 2016 took a 2,000 mu loan at 5% (due 2019), bought equipment for 5,371 mu (20-year life, 26 mu accum dep in 2016), technology licenses for 206 mu (5-year life), and financial shares for 636 mu. 2016 reflected a loss of 340 mu and cash of 1,476 mu.
In 2017: 5,000 clients paying 0.032 mu/mo subscription fees, plus 46 mu online sales (receivables at end of year: 164 mu). Purchased 680 mu materials (560 mu paid immediately, rest deferred; half consumed in sales). Staff costs 559 mu paid in full. Outside services 312 mu paid in full. Administrative costs 300 mu (40% paid, 60% deferred). Corporate tax rate 30% payable Q1 2018. Retain all profits to offset 2016 losses.

1. Prepare 2017 Income Statement.

2. Construct 31/12/2017 Balance Sheet.

3. What changes if a 20% dividend is paid at 24:00 on 31/12/2017?

4. What changes if a 20% dividend is approved before year-end to be paid on 30 June 2018?

EXAM DAY STRATEGY: DIVIDEND PAYMENT VS DIVIDEND DECLARATION

- **Paid on 31 Dec (Cash Outflow):** Cash decreases by dividend amount; Retained Earnings decrease by dividend amount. Total Assets and Total Equity contract.
- **Payable in the Future (Accrued Debt):** No cash leaves the company in 2017! Retained Earnings decrease, but a current liability (**Dividends Payable**) is created. Total Assets remain unchanged!
- **Prior Year Loss:** The 340 mu loss sits in Equity as a negative line item, offset by 2017's net profit.

Calculations & Statements

- **Revenues:** $5,000 \times 0.032 \times 12 + 46 = 1,920 + 46 = \mathbf{1,966.00}$ mu.
- **COGS:** Half of $680 = \mathbf{340.00}$ mu (Inventory $= 340.00$ mu).
- **Depreciation:** Equipment $(5,371/20 = 268.55)$ + Licenses $(206/5 = 41.20) = \mathbf{309.75}$ mu.
- **Operating Expenses:** $340 + 559 + 312 + 300 + 309.75 = \mathbf{1,820.75}$ mu.
- **EBIT, EBT & Net Income:** $\text{EBIT} = 1,966 - 1,820.75 = 145.25$ mu. Interest $= 5\% \times 2,000 = 100$ mu. $\text{EBT} = 45.25$ mu. Tax $(30\%) = 13.575$ mu. Net Income $= \mathbf{31.675}$ mu.
- **Ending Cash:** $1,476 + 1,802$ (collections) $- 560 - 559 - 312 - 120 - 100 = \mathbf{1,627.00}$ mu.

ASSETS at 31/12/2017	Amount (mu)	LIABILITIES & EQUITY at 31/12/2017	Amount (mu)
• Licenses (Net BV: 206 - 41.20)	164.800	• Subscribed Share Capital	6,003.000
• Technical Equipment (Net: 5,371 - 294.55)	5,076.450	• (-) Prior Year Loss (2016)	(340.000)
• Financial Shares in other companies	636.000	• Net Income Year 2017	31.675
TOTAL NON-CURRENT ASSETS	**5,877.250**	**TOTAL OWNERS' EQUITY**	**5,694.675**
• Inventories (Supplies)	340.000	• Long-term Bank Loan (due 2019)	2,000.000
• Accounts Receivable	164.000	• Accounts Payable (Suppliers)	120.000
• Cash & Bank Deposits	1,627.000	• Other Creditors (Admin expenses)	180.000
		• Taxes Payable (due Q1 2018)	13.575
TOTAL ASSETS	**8,008.250**	**TOTAL LIABILITIES & EQUITY**	**8,008.250**

DIVIDEND POLICY COMPARISON (QUESTIONS 3 & 4)

20% Dividend Amount: $20\% \times 31.675 = \mathbf{6.335}$ mu.

- **If Paid on 31/12/2017:** Cash drops to $1,620.665$ mu. Equity drops to $5,688.340$ mu. Total Balance $= 8,001.915$ mu.
- **If Payable on 30/06/2018:** Cash remains $1,627.000$ mu. Current Liabilities increase by $+6.335$ mu (**Dividends Payable**), Equity drops to $5,688.340$ mu. Total Balance remains balanced at **8,008.250 mu**!

Four-Phase Audit Trail (journal entries and ledger)

Phase 1: Hypotheses. The 31/12/2016 position is reconstructed from the data: capital 6,003; loan 2,000; equipment 5,371 less 26 accumulated depreciation; licenses 206; financial shares 636; cash 1,476; loss -340. Check: $1,476 + 5,345 + 206 + 636 = 7,663 = 2,000 + 5,663$. Licenses are assumed not amortized in 2016 (the statement gives no 2016 amortization) and are amortized over 5 years in 2017; equipment is depreciated over 20 years. Stocks were exhausted at end 2016, so half of the 680 purchases is consumed (COGS 340). Interest is paid at year end; tax is payable in Q1 2018.

Phase 2: Definitions. $A = L + E$ with $E = \text{Share capital} - \text{Prior-year loss} + \text{Net income}_{2017}$. Financial shares are a non-current investment, not an expense. A declared dividend reduces equity at once but reduces cash only when paid; in between it is a current liability.

Phase 3: Journal entries 2017 (m.u.). Each entry was balanced in Python (debits = credits).

#	Event	Debit	Credit
1	Opening balance 31/12/2016 (A = 7,663 = L 2,000 + E 5,663)	Cash 1,476.000 Technical equipment 5,371.000 Licenses 206.000 Financial shares 636.000 Prior-year loss 340.000	Accum. depreciation 26.000 Bank loan 2,000.000 Share capital 6,003.000
2	Revenue: 5,000 x 0.032 x 12 + 46 online sales (164 uncollected)	Cash 1,802.000 Accounts receivable 164.000	Revenue 1,966.000
3	Purchases of materials 680 (560 paid, 120 deferred)	Inventory 680.000	Cash 560.000 Accounts payable 120.000
4	Consumption: half of the purchases	COGS 340.000	Inventory 340.000
5	Staff costs paid	Staff 559.000	Cash 559.000
6	Supplies and outside services paid	Services 312.000	Cash 312.000
7	Administrative expenses 300 (40% paid, 60% deferred)	Administration 300.000	Cash 120.000 Other creditors 180.000
8	Depreciation equipment 5,371/20	Depreciation 268.550	Accum. depreciation 268.550
9	Amortization licenses 206/5	Amortization 41.200	Licenses 41.200
10	Interest 5% x 2,000 paid at year end	Interest 100.000	Cash 100.000
11	Corporate tax 30% x 45.25 (payable Q1 2018)	Tax expense 13.575	Taxes payable 13.575

Closing ledger account (31/12/2017)	Balance (m.u.)
Cash	1,627.000 Dr
Accounts receivable	164.000 Dr
Inventory	340.000 Dr
Technical equipment	5,371.000 Dr
Accum. depreciation	294.550 Cr
Licenses	164.800 Dr
Financial shares	636.000 Dr
Bank loan	2,000.000 Cr
Accounts payable	120.000 Cr
Other creditors	180.000 Cr
Taxes payable	13.575 Cr
Share capital	6,003.000 Cr
Prior-year loss	340.000 Dr

Income statement from the ledger: $1{,}966.000 - 340.000 = 1{,}626.000$; minus $1{,}480.750$ ($559.000 + 312.000 + 300.000 + 268.550 + 41.200$) gives EBIT $= 145.250$; minus interest 100.000 gives EBT $= 45.250$; minus tax 13.575 gives net income 31.675.

Requirement 3 (20% dividend paid at 24:00 on 31/12/2017): dividend $= 0.20 \times 31.675 = 6.335$. Entry: debit retained earnings 6.335, credit cash 6.335. Cash $1{,}627.000 - 6.335 = 1{,}620.665$; assets $8{,}001.915$; equity $5{,}688.340$; $5{,}688.340 + 2{,}313.575 = 8{,}001.915$. **Requirement 4** (decided in 2017, paid 30/06/2018): debit retained earnings 6.335, credit dividends payable 6.335. Assets unchanged ($8{,}008.250$); equity $5{,}688.340$; current liabilities $313.575 + 6.335 = 319.910$; total $5{,}688.340 + 2{,}000.000 + 319.910 = 8{,}008.250$.

Phase 4: Units and sanity checks. Currency: m.u. (as in the statement); three decimals are kept because 268.55, 41.20 and 13.575 are exact. $A = L + E$: $8{,}008.250 = 2{,}313.575 + 5{,}694.675$, and it still holds in both dividend variants. Cash identity: $1{,}476 + 1{,}802 - 560 - 559 - 312 - 120 - 100 = 1{,}627$. Limit check: paying at once shrinks both sides by 6.335; deferring only moves 6.335 from equity to liabilities. No official numerical key is available in `sources/` (only the statement PDF).

Study notes written from the official course material (T3_Additional_EXERCISES.pdf; Topic 3 Financial Management (I).pdf). Online version: subjects/business-management/problemas/topic-3-extra-exam-problems.html